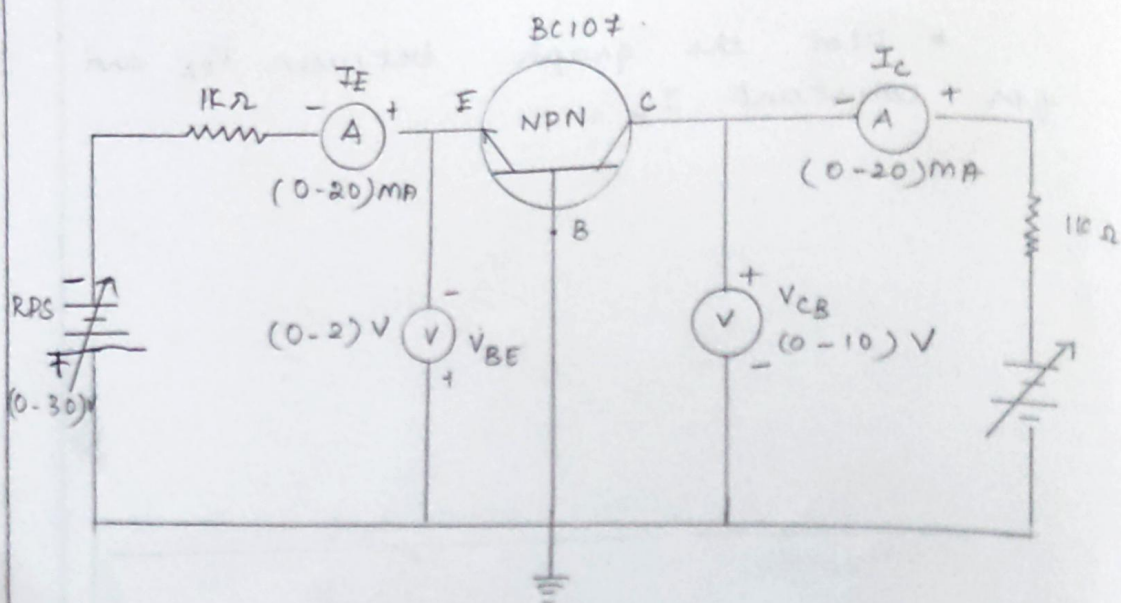
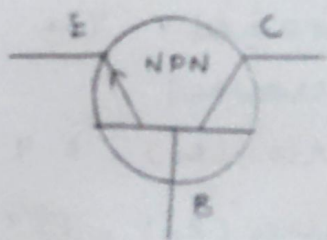


CIRCUIT DIAGRAM :

CB CONFIGURATION



Symbol



Aim:

To determine the input or output characteristics of BJT in common Base.

To determine the voltage gain, current gain, input or output resistance from the characteristics.

Apparatus / components / Device Required:

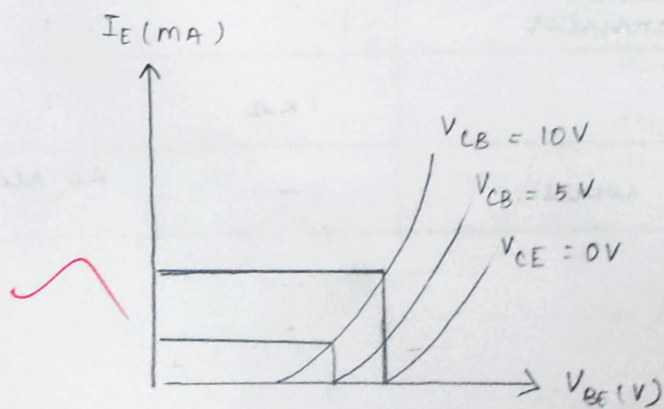
SL No.	Apparatus / components Name	Type / Range	Quantity.
1.	Dual Regulated power supply (RPS)	(0-30)V	1
2.	Voltmeter	(0-10)V	1
3.	Ammeter	(0-20)mA	2
4.	voltmeter	(0-2)V	1
5.	Bread Board	-	1
6.	Bipolar Junction Transistor		1
7.	Resistor	1K Ω	1
8.	connecting wires.	-	As required

TABULATION :

INPUT CHARACTERISTICS

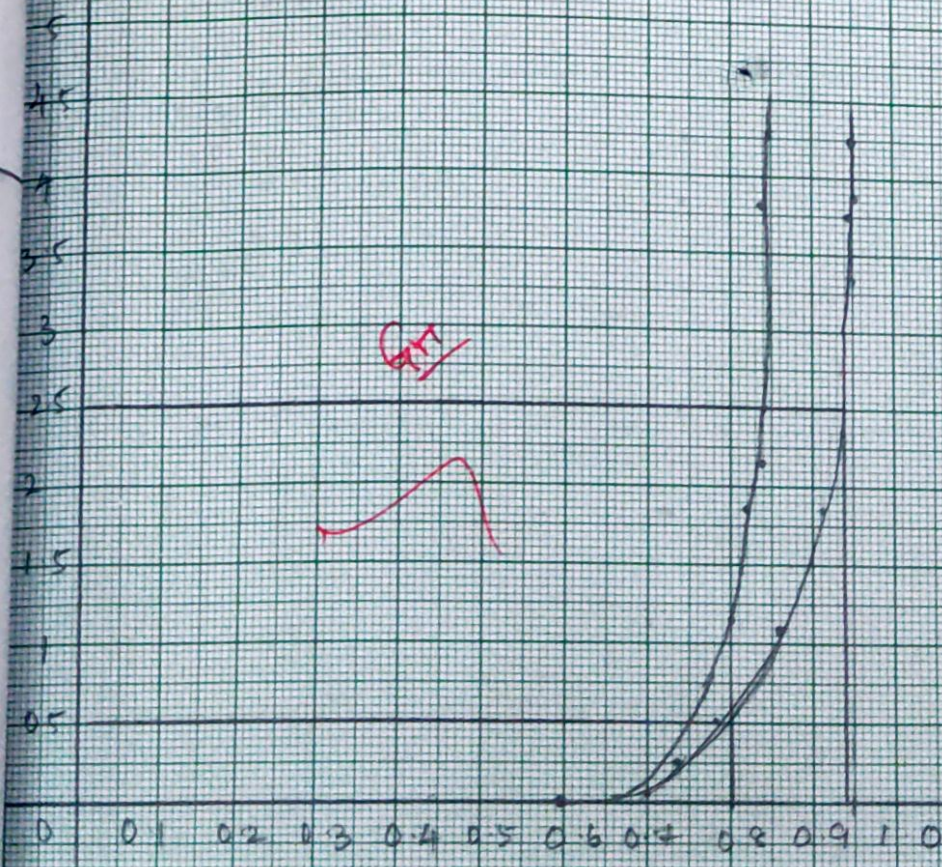
$V_{CB} = 0V$		$V_{CB} = 1V$	
$V_{BE} (V)$	$I_E (mA)$	$V_{BE} (V)$	$I_E (mA)$
0.62	0.07	0.71	0.15
0.74	0.26	0.73	0.20
0.78	0.54	0.78	0.54
0.87	1.16	0.80	1.12
0.91	1.68	0.82	1.72
0.93	2.36	0.83	2.28
0.94	2.81	0.83	2.86
0.94	3.35	0.84	3.30
0.94	3.76	0.84	3.81
0.94	4.2	0.84	4.36

MODEL GRAPH



Input characteristics

Scale

X-axis $1\text{cm} = 0.1\text{V}$ Y-axis $1\text{cm} = 0.5\text{mA}$ 

input impedance :-

$$R_i = \frac{\Delta V_{BE}}{\Delta I_E}$$

$$= \frac{0.94 - 0.79}{2.5 - 0.5 \times 10^{-3}}$$

$$= 0.075 \times 10^3 \Omega$$

Lab No:

CB

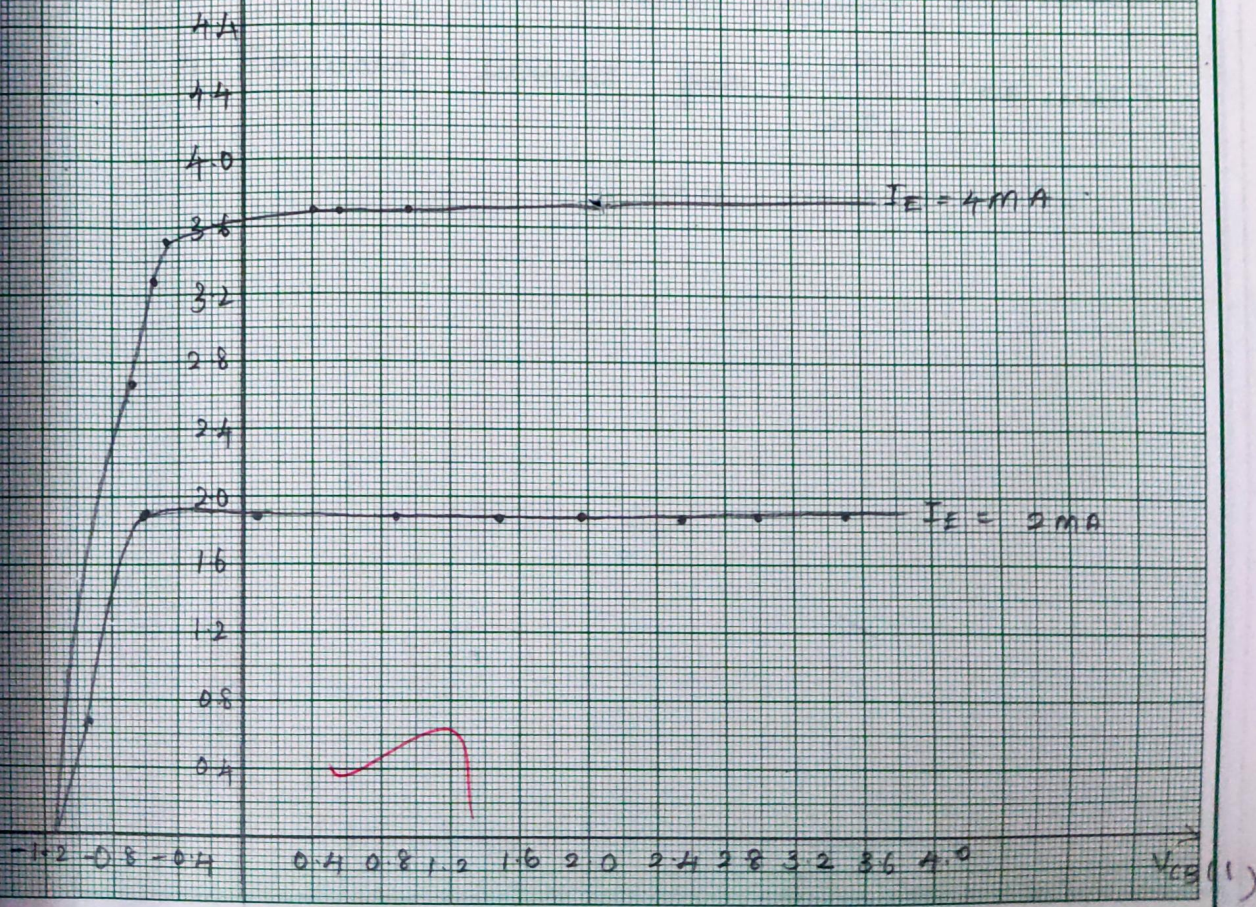
Date :

I_E (mA)

Scale

X axis 1 cm = 0.4 V

Y axis 1 cm = 0.4 mA



Formula:

1. Input resistance $R_i = \frac{V_{BE}}{I_E} (\Omega)$

with constant I_B .

2. Output resistance $R_o = \frac{\Delta I_c}{\Delta I_E}$

with $\Delta = \frac{\Delta I_c}{\Delta I_E}$ with I_B constant.

3. Reverse voltage gain $= \frac{\Delta V_{BE}}{\Delta V_{CB}}$ (no unit)

with constant ΔI_E constant.

4. Forward current gain $= \frac{\Delta I_c}{\Delta I_E}$ (no unit)

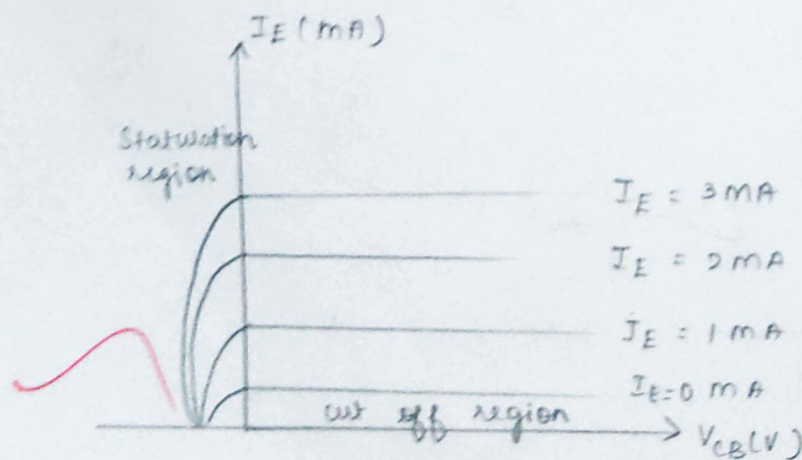
with constant ΔV_{CB} .



OUTPUT CHARACTERISTICS

$I_E = 3 \text{ mA}$		$I_E = 4 \text{ mA}$	
$V_{CB} (V)$	$I_C (mA)$	$V_{CB} (V)$	$I_C (mA)$
-0.68	0.62	-0.71	1.29
-0.51	1.91	-0.73	1.76
-0.38	1.91	-0.69	2.20
0.27	1.92	-0.68	2.69
0.83	1.93	-0.66	3.23
1.47	1.93	-0.63	3.54
1.98	1.93	-0.40	3.88
2.58	1.93	0.01	3.88
3.08	1.93	0.54	3.88
3.52	1.93	1.08	3.88

MODEL GRAPH:



output characteristics

Procedure :

Input characteristics :

* Connections are made as per the circuit diagram.

* For plotting the input characteristics, the output voltage V_{CE} is kept constant at 0V and for different values of V_{CE} note down the values of I_E .

* Repeat the above step keeping V_{CB} at 4V and 6V. All the readings are tabulated.

* A graph is drawn between V_{EB} and I_E for constant V_{CB} .

Output characteristics :-

* Connections are made as per the circuit diagram.

* For plotting the output characteristics, the input I_E is kept constant at 10mA and for different values of V_{CB} , note down the values of I_C .

* Repeat the above step for the value of I_E at 20mA, 40mA, and 60mA, all the readings are tabulated.



* A graph is drawn between V_{CB} and I_c for constant I_E .

RESULT:

Thus the input and output characteristics of a transistor in common Base mode is obtained. The required parameters are determined.

$$\text{Input impedance (R}_i\text{)} = 0.075 \times 10^3 \Omega.$$